

Wave Intensity (Cambridge (CIE) A Level Physics): Revision Note



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Wave intensity

- **Progressive waves** transfer **energy**
- The amount of energy passing through a unit area per unit time is the **intensity** of the wave
- Therefore, the **intensity** is defined as **power per unit area**

$$I = \frac{P}{A}$$

- Where
 - I is intensity in W m^{-2}
 - P is power in W
 - A is area in m^2
- The **area** the wave passes through is **perpendicular** to the direction of its **velocity**
- The **intensity** of a progressive wave is also **proportional** to its **amplitude squared** and frequency squared

$$I \propto A^2$$

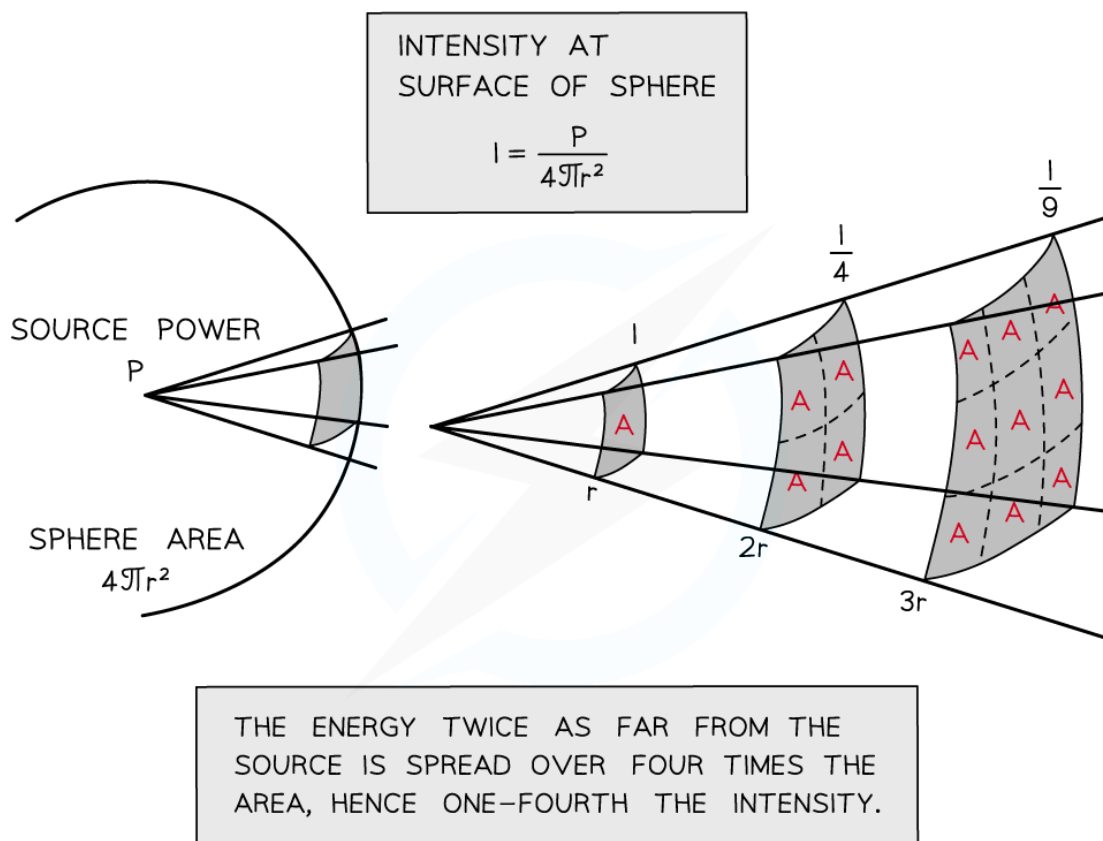
$$I \propto f^2$$

- Where
 - A is amplitude in m
 - f is frequency in Hz
- Recall that $\propto$ means "proportional to"
- This means that if the frequency or the amplitude is doubled, the intensity increases by a factor of 4 (2^2)

Spherical waves

- A spherical wave is a wave from a point source which spreads out equally in all directions
- The area the wave passes through is the **surface area** of a sphere: $4\pi r^2$
- As the wave travels further from the source, the energy it carries passes through increasingly larger areas as shown in the diagram below:

Inverse square law for intensity



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Intensity is proportional to the amplitude squared

- Assuming none of the wave energy is absorbed, the intensity I decreases with increasing distance from the source
- The **inverse square law** states that

$$I \propto \frac{1}{r^2}$$

- The inverse square law means when the source is twice as far away, the intensity is 4 times smaller



Worked Example

The intensity of a progressive wave is proportional to the square of the amplitude of the wave. It is also proportional to the square of the frequency. The variation with time t of displacement x of particles when two progressive waves Q and P pass separately through a medium, are shown on the graphs.

The intensity of wave Q is I_0 . What is the intensity of wave P?

Answer:

Step 1: Compare waves Q and P:

- The amplitude of wave **Q** is twice the amplitude of wave **P**
- The frequency of wave **P** is double the frequency of wave **Q**

Step 2: Evaluate how amplitude affects the intensities:

- **Q** has twice the amplitude of **P**

$$\frac{A_Q}{A_P} = 2$$

- The ratio of amplitudes squared is equal to the ratio of intensities, because intensity and amplitude squared are proportional

$$\frac{A_Q^2}{A_P^2} = 4 = \frac{I_Q}{I_P}$$

$$I_Q = 4I_P$$

- So **Q**'s larger amplitude increases its intensity by a factor of 4

Step 3: Evaluate how frequency affects the intensities:

- **Q** has half the frequency of **P**

$$\frac{f_Q}{f_P} = \frac{1}{2}$$

- The ratio of frequencies squared is equal to the ratio of intensities because intensity and frequency squared are proportional

$$\frac{f_Q^2}{f_P^2} = \frac{1}{4} = \frac{I_Q}{I_P}$$

$$I_Q = \frac{1}{4}I_P$$

- So **Q**'s lower frequency decreases its intensity by a factor of 4

Step 4: Determine the intensity of wave **P** in terms of I_0 :

- **P** has a lower amplitude which reduces its intensity by a factor of 4 compared to **Q**
- However, **P** has a higher frequency which increases its intensity by a factor of 4 compared to **Q**
- These effects **cancel out**, so wave **P** has an intensity of I_0 , which is equal to that of wave **Q**



Examiner Tips and Tricks

The key concept with intensity is that it has an inverse square relationship with distance (not a linear one). This means the energy of a wave decreases very rapidly with increasing distance